5Sense — Software Architecture Flowchart

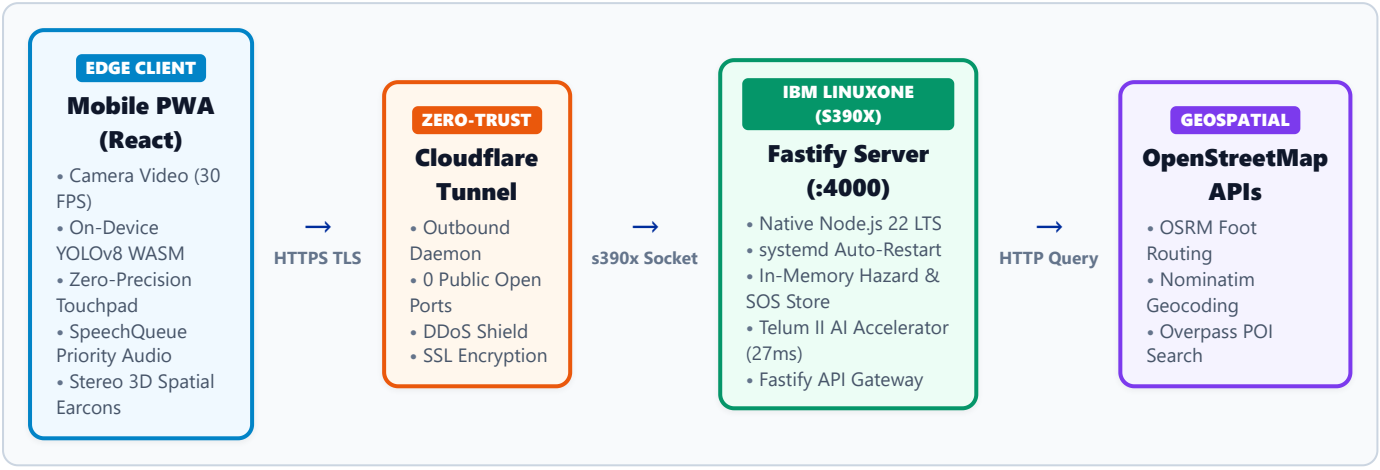
Clean Technical Code Flow & System Component Interactions

IBM Z DATATHON 2026

Host: IBM LinuxONE 5 (z17 s390x) • **Live:** nickel-highlight-hide-units.trycloudflare.com

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1. System Architecture Flowchart (End-to-End Edge to Mainframe Connection)



2. Code Execution Flowchart (How Code Responds to User Gestures in NavigatePage.tsx)

1-TAP (<240MS) • VISION

Instant Path Scan

1. User Touch Release
`handlePadPressEnd()` checks tap duration < 240ms → triggers `handleSimpleDetection()`.

↓

2. Neural Inference (<10ms)
`YoloDetector.ts` / `ObjectDetector.ts` runs on-device WASM tensor detection on camera frame.

↓

3. Geometric Distance
`estimateDistance()` computes physical distance in metres from bounding box dimensions.

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4. Auditory Feedback
If obstacle < 2.5m → `playSpatialHazardTone()` + speaks alert.
If clear → `playClearPathChime()` + speaks "*Path clear*".

2-TAP HOLD (>240MS) • NAV

Push-to-Talk Routing

1. Hold Detection
`handlePadPressStart()` holds > 240ms → triggers `speechQueue.setMicActive(true)`.

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2. Instant Audio Silence
SpeechQueue kills all background turn chimes and speech in 0ms so mic never records app's own sound.

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3. Spoken Voice Query
User releases finger → Web Speech API records destination → sends to LinuxONE `/api/geocode/search`.

↓

4. Turn-by-Turn Launch
LinuxONE computes OSRM foot route → launches `VoiceCoPilot.tsx` turn guidance ("*Take right*" / "*Take left*").

3-TAP / SHAKE • SOS

Emergency SOS Broadcast

1. SOS Gesture Trigger
3 rapid taps or vigorous phone shake (`useShakeDetection()`) triggers `handleEmergencySos()`.

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2. Alarm & Haptic Pattern
`playSosEmergencyAlarm()` sounds alternating siren + pulsed high-intensity SOS vibration.

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3. Mainframe Session Dispatch
Sends live GPS coordinates to LinuxONE `/api/sos/start` → creates persistent session token.

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4. Reverse Geocoded Broadcast
Backend resolves street address → trusted contacts receive emergency SMS with live GPS tracking link.

5Sense — Navigation & Priority Speech Engine

Voice Co-Pilot Walking Radar, 3D Spatial Audio & Priority-Interrupt Queue

VOICE CO-PILOT FLOW

Architecture: Edge Audio • LinuxONE Backend

3. Voice Co-Pilot Turn Guidance Flowchart (3-Stage Walking Distance Radar & 3D Binaural Earcons)

As the pedestrian walks, `VoiceCoPilot.tsx` tracks live GPS coordinates (`navigator.geolocation.watchPosition`) and computes remaining distance to the upcoming turn corner:

20 Metres
Preparation Notice

Audio: `playSpatialTurnCue()` pans ascending chime to turning ear.
Voice: *"In 20 metres, take right onto Market Street."*
Haptic: Single 50ms vibration pulse.

8 Metres
Prompt Notice

Audio: Mid-frequency waypoint ping panned to turning ear.
Voice: *"In a few steps, take right."*
Haptic: Double 40ms vibration pulse.

2 Metres (Corner)
Turn Execution

Audio: Dual melodic chime in turning ear.
Voice: *"Take right now onto Market Street."*
Advance: Steps to next turn: *"Then, in 45m, take left."*

Compass Veering Detection: Device magnetometer (`webkitCompassHeading`) is continuously compared to expected step bearing. If user drifts >35° off path for >3s → announces: *"Veering right. Turn slightly left to stay on path."*

4. SpeechQueue Priority-Interrupt Flowchart (Strict Collision-Free Audio Engine in SpeechQueue.ts)

Priority Hierarchy

1. CRITICAL: Urgent camera obstacles (<2.5m) & SOS alerts. Interrupts everything immediately.
2. NAVIGATION: Turn instructions (*"Take right"*, *"Take left"*).
3. INFO / AMBIENT: Nearby places, status checks.

⇔

Hardware Microphone Lockout Rule

The exact millisecond user touches screen to speak → `speechQueue.setMicActive(true)` is invoked.
- `window.speechSynthesis.cancel()` kills active speech in 0ms.
- All incoming background audio alerts are dropped or queued.
- Zero audio collision: navigation never talks over user voice.

5. Software Module Interaction Summary

Module File	Role in Flow	Input → Output
NavigatePage.tsx	Central gesture & camera coordinator	Touch events (1, 2, 3 taps) → Triggers detection, voice search, or SOS
ObjectDetector.ts / YoloDetector.ts	On-device WASM neural inference	640×480 video frame → Bounding boxes, class label & distance in metres
SpeechQueue.ts	Priority-interrupt audio manager	Spoken text + Priority → SpeechSynthesis with mic silence lockout
VoiceCoPilot.tsx	Turn-by-turn walking guidance	GPS lat/lon → 3-stage radar (20m, 8m, 2m) with 3D left/right ear cues
backend/src/server.ts	LinuxONE Fastify REST gateway	Geocode query → OSRM foot path with "Take right", "Take left" instructions